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1. EXPLANATION AND REASON OF STRATEGY:

The indication which i use in my strategy is:

1. SMA (Simple Moving Average)

Think of SMA as the **average closing price of a stock over a set number of days**, like 30 or 90, which i use in my strategy.

It smooths out the daily ups and downs so you can see the overall trend more clearly

 **Why it is useful (reason):** It helps investors identify if a stock is generally trending up, down, or sideways.

2. RSI (Relative Strength Index)

RSI is like a **speedometer for a stock's price movements** — it tells you if it's moving too fast and might need to slow down and also for identification of revision of trend.

It measures how **"overbought"** (too many buyers, might fall soon) or **"oversold"** (too many sellers, might rise soon) a stock is, on a scale of 0 to 100.

Above 70 = might be too hot.

Below 30 = might be too cold.

 **Why it is useful (reason):** Helps spot when a trend might be reversion..

3. EMA (Exponential Moving Average)

EMA is like SMA's smarter cousin. It also tracks average prices, but it **gives more weight to recent prices**.

So if something just changed in the market, the EMA reacts faster than the SMA

 **Why it is useful (reason):** Traders use it for faster signals to enter or exit a trade.

4. ADX (Average Directional Index)

ADX doesn't care if prices are going up or down — it just tells you **how strong the trend is**.

It's like checking how strong the wind is, not the direction it's blowing. A reading above 25 means the trend (up or down) is strong. Below that? The market might just be wandering aimlessly.

 **Why it is useful (reason):** It helps you avoid trading in choppy, no-direction zones.

Absolutely! Let's break down **DI+ (Plus Directional Indicator)** and **DI- (Minus Directional Indicator)** in a **human-friendly** way, continuing with the finance context.

DI+ (Plus Directional Indicator)

Think of **DI+** as a tracker for **upward strength**.

It measures **how strongly buyers are pushing prices upward**.

 If **DI+** is high, it means the price is making higher highs — bulls (buyers) are in control, and the stock is climbing with force.

DI- (Minus Directional Indicator)

Now, **DI-** is just the opposite — it measures **how strongly sellers are pushing prices downward**.

 If **DI-** is high, it means the price is making lower lows — bears (sellers) are in control, and the stock is falling with momentum.

- When **DI+ is above DI-**, the bulls are stronger → **possible uptrend**.
- When **DI- is above DI+**, the bears are stronger → **possible downtrend**.
- When they're close together or crisscrossing a lot, the market might be **sideways or indecisive**.

Why it is useful (reason) :

Traders often use the **crossover between DI+ and DI-** as a signal to buy or sell. Combined with **ADX** (which tells you the strength of the trend), you get a fuller picture:

- **ADX high + DI+ > DI-** = strong uptrend.
 - **ADX high + DI- > DI+** = strong downtrend.
-

5. Bollinger Bands

Imagine a stock's price walking a tightrope. The Bollinger Bands are **rubber bands around the price**, showing how wild or calm the market is.

There's a middle line (a moving average), and two bands that expand or contract based on price volatility.

When the bands are tight = calm market.

When they're wide = high volatility.

 **Why it is useful (reason):** If prices hit the upper or lower band, it might be a sign of reversal or breakout.

I use this indicator on basis of above mention reason

Now it's time for strategy explanation

Code use for downloading data of indian stokes

```
df = yf.download('jmfinancil.NS', start='2023-01-03', end='2025-5-20', auto_adjust=False)
if df.empty:
    print("No data available. Exiting.")
    exit()
df.columns = df.columns.get_level_values(0)
df.dropna(inplace=True)
df = df[['Open', 'High', 'Low', 'Close', 'Volume']]
df.index.name = 'Date'
```

Overall Strategy Structure

This is a trend-based and mean-reversion based trading strategy using:

- **ADX** for trend strength
 - **DI+ and DI- crossovers** for trend direction
 - **RSI** and **SMA crossovers** for momentum
 - **Bollinger Bands** (price vs upper/lower bands) for volatility signals
 - Basic **stop-loss (SL)** and **take-profit (TP)** risk controls
-

1. If $ADX > 25 \rightarrow$ A strong trend exists

Within this strong trend, we check who's stronger: **buyers (DI+)** or **sellers (DI-)**.

If DI+ crosses above DI- (bullish signal):

- If we don't already have a position, **buy**.
- If we already have a position (could be sell or buy), **close it and re-enter a buy** (maybe reversing a short).

If DI- crosses above DI+ (bearish signal):

- If we don't have a position, **sell**, with a **take-profit at 90% of the current price** (expecting a 10% drop).

- If we already have a position, close it and sell again.

```
if (ADX>25) :
    if((crossover(self.DIP, self.DIN))):
        if(not(self.position)):
            self.buy()
        elif((self.position)):
            self.position.close()
            self.buy()

    elif (crossover(self.DIN, self.DIP)):
        if(not(self.position)):
            # tp=price*0.9
            self.sell()
        elif((self.position)):
            self.position.close()
            self.sell()
```

If $ADX < 20$ → Weak or no trend

- Close any existing position — no clear trend means it's risky to stay in the market.

```
self.sell()
elif(ADX<20):
    if (self.position):
        self.position.close()
```

Other Situations — Independent Signals

These rules apply **regardless of ADX**, and look for **momentum or reversal** signals using RSI, SMA, and Bollinger Bands.

▲ If price crosses above the upper Bollinger Band OR RSI crosses above its upper bound:

- **Sell** — the stock may be overbought or overextended.

```
if crossover(self.upper,price)|crossover (self.rsi,self.upper_bound_RSI):  
    self.sell()
```

▼ If RSI crosses above its lower bound OR (short SMA crosses above long SMA AND RSI is not overbought) OR price crosses above the lower Bollinger Band:

- **Buy**, with:
 - **Stop-loss** set at 95% of the current price (in case it drops)
 - **Size** of 0.8 (not full position, likely a more cautious entry)

```
elif crossover(self.lower_bound_RSI,self.rsi) or ((crossover(self.short_sma,self.long_sma1)&(self.rsi<self.upper_bound_RSI))|crossover(price,self.lower)):  
    sl=price*0.95  
    size=,8  
    self.buy(size=size,sl=sl)
```

🔄 If long SMA crosses above short SMA AND RSI is still not overbought:

- **Close the position** — this may signal the bullish momentum is fading.

```
self.buy(size=size,sl=sl)  
elif (crossover(self.long_sma1,self.short_sma)&(self.rsi<self.upper_bound_RSI)):  
    self.position.close()
```

🧠 Summary :

- **Only trade when there's a strong trend** (ADX > 25).
 - Buy if buyers are gaining control.
 - Sell if sellers are gaining control.
 - **Exit trades if there's no clear trend** (ADX < 20).
 - Use **RSI, SMA, and Bollinger Bands** for additional buy/sell signals — especially in volatile or overbought/oversold conditions.
 - **Protect trades** with stop-loss (for buys) and take-profit (for sells).
-

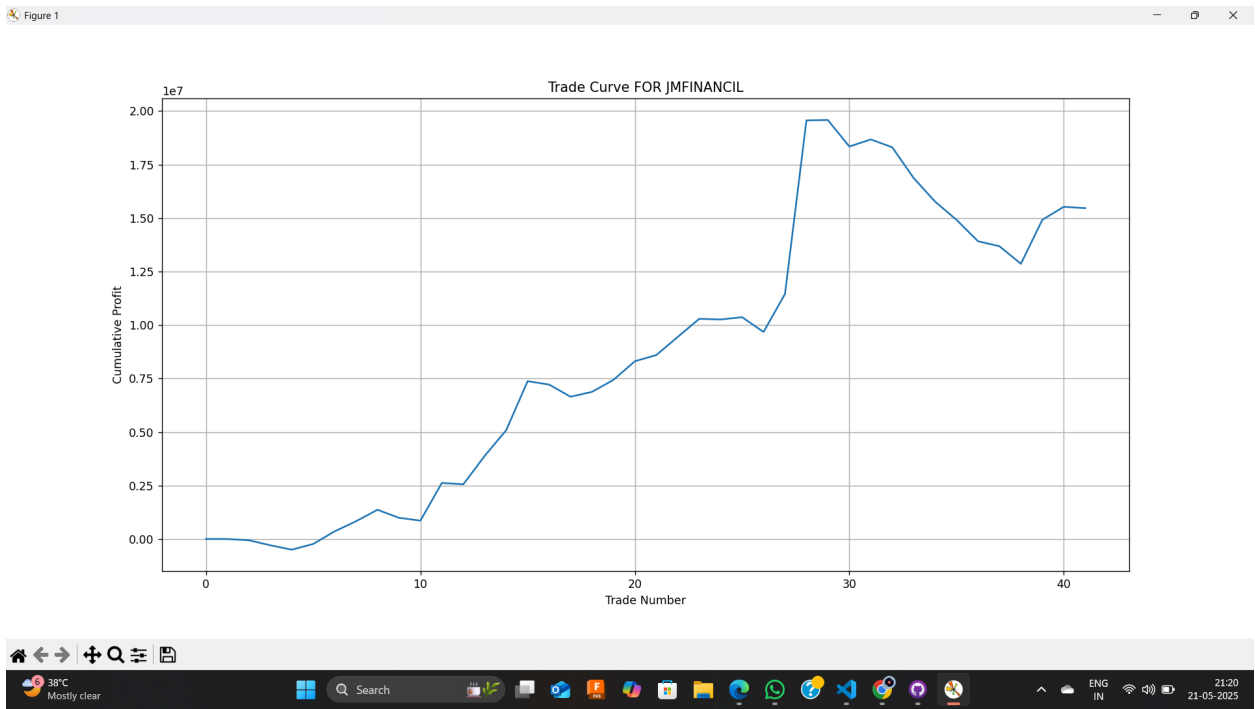
RESULTS

As i tested this strategy over jmfinancil so

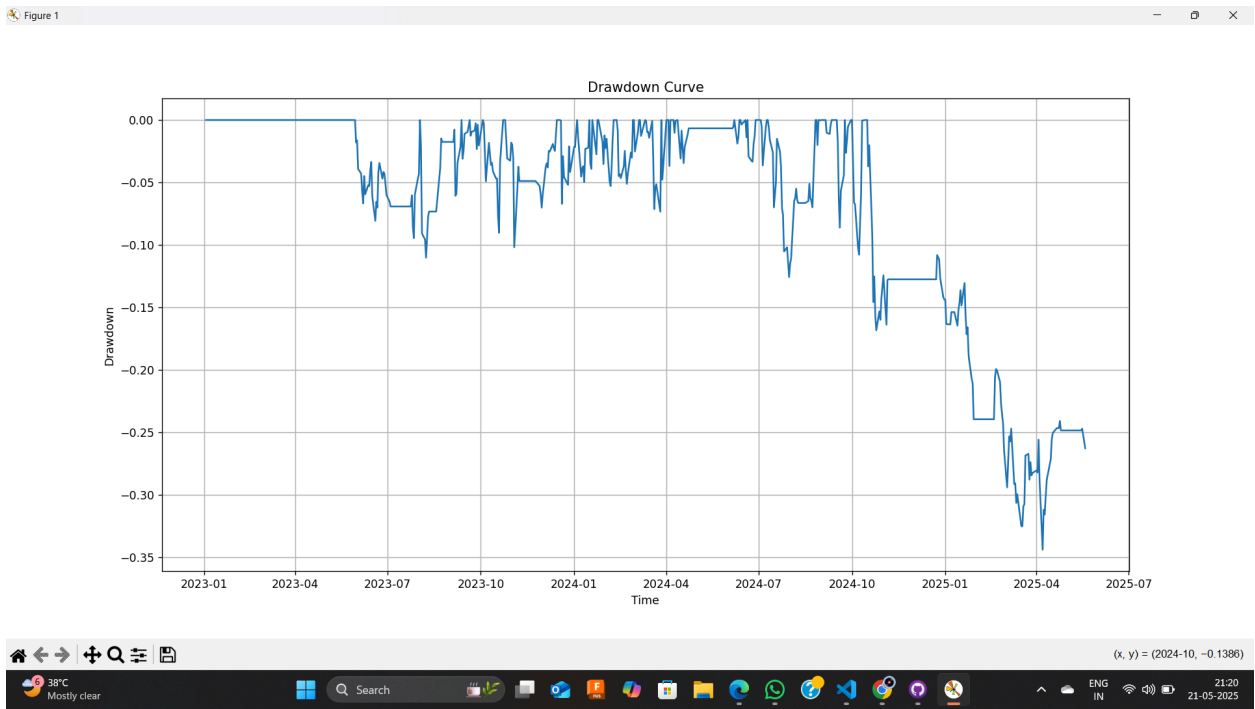
1.Equity curve for JMFINANCIL



2. Trade curve for JMFINANCIL



3. Drawdown curve for JMFINANCIL



4.. Curve for JMFINANCIL



5. Performance Metrics and trade entry and exit data

```
warnings.warn(  
Start                2023-01-03 00:00:00  
End                  2025-05-19 00:00:00  
Duration              867 days 00:00:00  
Exposure Time [%]    60.20583  
Equity Final [$]     24978796.4687  
Equity Peak [$]      33887236.02274  
Return [%]           149.78796  
Buy & Hold Return [%] 84.18168  
Return (Ann.) [%]    48.54194  
Volatility (Ann.) [%] 52.83569  
CAGR [%]             30.48396  
Sharpe Ratio         0.91873  
Sortino Ratio        2.32371  
Calmar Ratio         1.4109  
Alpha [%]            125.75838  
Beta                 0.28545  
Max. Drawdown [%]    -34.4049  
Avg. Drawdown [%]    -5.64692  
Max. Drawdown Duration 215 days 00:00:00  
Avg. Drawdown Duration 22 days 00:00:00  
# Trades              42  
Win Rate [%]         52.38095  
Best Trade [%]       37.78644  
Worst Trade [%]      -7.65634  
Avg. Trade [%]       3.25599  
Max. Trade Duration  71 days 00:00:00  
Avg. Trade Duration  16 days 00:00:00  
Profit Factor        3.30262  
Expectancy [%]       3.65861  
SQN                  1.59718  
Kelly Criterion      0.3228  
_strategy            my_strategy  
_equity_curve        ...  
_trades              Size Entr...  
dtype: object
```

dtype: object

	Size	EntryBar	ExitBar	EntryPrice	ExitPrice	SL	...	Entry_λ(λ(λ(DM(df...))	Exit_λ(λ(λ(DM(df...))	Entry_λ	Exit_λ	Entry_λ(λ)	Exit_λ(λ)
0	-6147	97	100	72.000000	70.500000	NaN	...	51.373594	58.623924	65.202500	66.405000	59.783965	61.088599
1	-5703	102	116	72.250000	72.449997	NaN	...	62.528254	40.696195	67.405000	72.232500	62.013885	68.932217
2	-132741	97	116	72.000000	72.449997	NaN	...	51.373594	40.696195	65.202500	72.232500	59.783965	68.932217
3	-137313	116	124	72.449997	74.150002	NaN	...	40.696195	19.403589	72.232500	73.055000	68.932217	71.004865
4	-133355	137	139	72.449998	74.400002	NaN	...	29.150445	29.027254	73.839999	74.032499	71.908925	72.511324
5	127798	139	150	74.400002	76.500000	NaN	...	29.027254	20.992285	74.032499	75.532500	72.511324	70.336115
6	-117648	155	158	83.099998	78.099998	NaN	...	22.278327	15.805762	76.242500	77.077500	69.849635	71.152487
7	-115228	166	180	89.949997	85.849998	NaN	...	31.090579	29.136755	78.592500	86.180000	68.968555	76.623772
8	-126234	180	199	85.849998	81.599998	NaN	...	29.136755	29.866446	86.180000	86.407500	76.623772	80.790020
9	-139384	199	209	81.599998	84.300003	NaN	...	29.866446	17.330817	86.407500	85.212500	80.790020	78.240007
10	-131155	221	223	83.849998	84.849998	NaN	...	25.342949	26.536841	83.415000	83.615000	80.108532	80.363751
11	128064	223	250	84.849998	98.599998	NaN	...	26.536841	38.462088	83.615000	97.830000	80.363751	88.902314
12	-128064	250	252	98.599998	99.099998	NaN	...	36.959559	36.959559	97.830000	98.860000	88.902314	92.864861
13	126772	252	270	99.099998	109.599998	NaN	...	36.959559	32.929646	98.860000	106.990000	92.864861	97.967362
14	-51003	270	290	109.599998	86.099998	NaN	...	32.929646	46.398896	106.990000	99.245000	97.967362	89.077822
15	-75769	270	292	109.599998	79.400002	NaN	...	32.929646	45.967652	106.990000	97.090000	97.967362	81.375153
16	38968	292	293	79.400002	75.335003	NaN	...	45.967652	47.724772	97.090000	95.805000	81.375153	76.845349
17	181171	295	298	76.050003	72.912500	NaN	...	52.276456	57.549444	93.102500	88.930000	71.467501	65.175389
18	37009	302	308	72.000000	78.150002	68.352497	...	71.593745	51.640796	83.309999	77.720000	62.144695	69.186141
19	177644	298	308	75.000000	78.150002	70.489997	...	57.549444	51.640796	88.930000	77.720000	65.175389	69.186141
20	223178	308	318	78.150002	82.050003	NaN	...	51.640796	17.260986	77.720000	79.405001	69.186141	72.963612
21	43137	348	354	78.199997	84.900002	NaN	...	25.939390	20.486295	79.982500	80.097000	76.102962	75.393513
22	82812	348	365	78.199997	88.449997	NaN	...	25.939390	35.988885	79.982500	82.745499	76.102962	74.367715
23	61383	348	366	78.199997	91.949997	72.912500	...	25.939390	39.723042	79.982500	83.528500	76.102962	75.272409
24	-13718	370	392	96.800003	98.989998	NaN	...	59.453816	16.006905	86.931999	99.114500	76.224765	92.321478
25	-104755	369	392	100.000000	98.989998	NaN	...	54.394723	16.006905	86.088499	99.114500	75.346411	92.321478
26	-97936	366	392	91.949997	98.989998	NaN	...	39.723042	16.006905	83.528500	99.114500	75.272409	92.321478
27	170851	397	403	92.150002	102.550003	86.240999	...	21.423106	29.925490	98.328499	97.404500	89.700971	87.566891
28	209232	403	453	102.550003	141.300003	NaN	...	29.925490	17.932329	97.404500	145.151001	87.566891	127.276910
29	-6826	485	486	127.500000	125.000000	NaN	...	44.221862	42.186538	136.821500	136.383500	127.085231	125.397286
30	-225052	485	492	127.500000	133.000000	NaN	...	44.221862	22.075226	136.821500	133.866999	127.085231	122.011783
31	184951	494	495	122.599998	124.389999	114.950000	...	19.536684	19.896531	131.977500	119.616937	118.576491	
32	201226	498	498	114.000000	112.176002	NaN	...	24.826716	24.826716	127.737000	127.737000	114.504735	114.504735
33	197783	499	508	114.500000	107.235997	NaN	...	26.459503	43.124073	126.589499	119.873500	113.724455	105.917077
34	199507	509	510	107.750000	102.219999	NaN	...	45.826920	49.523370	118.727999	117.660500	104.512798	103.820429
35	213376	526	532	96.610001	92.615498	NaN	...	36.098299	23.667950	106.437500	103.605500	96.525410	91.523459
36	215975	533	533	92.290001	87.675501	NaN	...	25.925392	25.925392	102.524000	102.524000	89.864142	89.864142
37	42961	541	542	89.080002	83.618997	NaN	...	24.090143	28.347520	96.315000	95.406499	86.531410	84.613189

17	2024-03-15	2024-03-20	76.050003	72.912500	-5.684246e+05	-0.041256
18	2024-03-27	2024-04-05	72.000000	78.150002	2.276054e+05	0.085417
19	2024-03-20	2024-04-05	75.000000	78.150002	5.595789e+05	0.042000
20	2024-04-05	2024-04-23	78.150002	82.050003	8.703945e+05	0.049904
21	2024-06-06	2024-06-14	78.199997	84.900002	2.890181e+05	0.085678
22	2024-06-06	2024-07-02	78.199997	88.449997	8.488230e+05	0.131074
23	2024-06-06	2024-07-03	78.199997	91.949997	8.440162e+05	0.175831
24	2024-07-09	2024-08-09	96.800003	98.989998	-3.004235e+04	-0.022624
25	2024-07-08	2024-08-09	100.000000	98.989998	1.058028e+05	0.010100
26	2024-07-03	2024-08-09	91.949997	98.989998	-6.894695e+05	-0.076563
27	2024-08-19	2024-08-27	92.150002	102.550003	1.776851e+06	0.112859
28	2024-08-27	2024-11-06	102.550003	141.300003	8.107740e+06	0.377864
29	2024-12-24	2024-12-26	127.500000	125.000000	1.706500e+04	0.019608
30	2024-12-24	2025-01-03	127.500000	133.000000	-1.237786e+06	-0.043137
31	2025-01-07	2025-01-08	122.599998	124.389999	3.310625e+05	0.014600
32	2025-01-13	2025-01-13	114.000000	112.176002	-3.670359e+05	-0.016000
33	2025-01-14	2025-01-27	114.500000	107.235997	-1.436696e+06	-0.063441
34	2025-01-28	2025-01-29	107.750000	102.219999	-1.103274e+06	-0.051323
35	2025-02-19	2025-02-28	96.610001	92.615498	-8.523310e+05	-0.041347
36	2025-03-03	2025-03-03	92.290001	87.675501	-9.966166e+05	-0.050000
37	2025-03-13	2025-03-17	89.080002	83.618997	-2.346102e+05	-0.061305
38	2025-03-04	2025-03-17	90.000000	86.146000	-8.194066e+05	-0.042822
39	2025-03-19	2025-03-24	88.550003	98.500000	2.055381e+06	0.112366
40	2025-03-24	2025-04-21	98.500000	100.900002	6.071932e+05	0.024365

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